

FC Card and Tag Security Enhancement Framework Using Hybrid Detection

NFC Hybrid Behavioral Anomaly Detection

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1. Decleration

1. Decleration

I hereby declare that this proposal is my original work. To the best of my knowledge, it has not been submitted previously, either in whole or in part, for the award of a degree or diploma at any university or higher education institution. I also confirm that it does not contain any previously published material or work written by another person, except where the relevant sources have been properly acknowledged and cited in the text.

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2. Abstract

The estimated budget for Novelty 4 is broken down into the following required research hardware, test media, software, and development resources. The price of Proxmark3 is LKR 25,000 and it is used to measure the NFC protocol and timing. The price of Flipper Zero is LKR 80,000 and it allows comparisons between the authorized genuine and emulated data. It is estimated that the cost of NFC cards for repeatable experiments is LKR 10,000. Python, Pandas and NumPy are free, scikit-learn, Jupyter Notebook, Proxmarket3 Client, and Proxmarket3 Firmware (Iceman) are also free. Human effort is not monetized as it is when data collection, analysis and documentation by the student are done.

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4. List of Abbreviations

Abbreviation	Meaning
NFC	Near Field Communication
ML	Machine Learning
AI	Artificial Intelligence
IDS	Intrusion Detection System
UID	Unique Identifier
RF	Radio Frequency
IF	Isolation Forest
API	Application Programming Interface
CSV	Comma-Separated Values
FPR	False Positive Rate
PR-AUC	Precision–Recall Area Under Curve
F1-score	Harmonic Mean of Precision and Recall
N4	Novelty 4
GUI	Graphical User Interface
JSON	JavaScript Object Notation

5. Introduction

Near Field Communication is a wireless technology that enables two devices or cards to communicate over a short range if they are compatible. NFC can be used in various practical applications such as access control, attendance management, identity management, transport system, payment system and smart authentication. In typical NFC-based systems, the credential is normally verified by determining whether the card or tag has been recognized, and whether the authentication process has been successful. But establishing a credential's validity does not necessarily imply that it is being used normally.

Unusual credential use may occur as a result of credential sharing, unauthorized credential access, multiple scans, an unusual movement of credential from location to location, or other unusual activity. For instance, a credential might be checked at the main entrance, and then checked at the Laboratory reader a few seconds later. Another certificate can generate numerous images in a really brief amount of time. These patterns could be indicative of misuse or unusual activity, even if the credential is accepted by the NFC reader. For the detection of well-defined suspicious patterns, traditional rule-based systems are useful. But they might not detect any new or subtle patterns which are not present in the rules. On the other hand, machine-learning models can detect atypical patterns based on behavioural data but they can be hard to explain and can produce false alarms if applied without domain expertise. Hence Novelty 4 suggests a hybrid system that merges the advantages of both systems.

The proposed system applies a set of predefined rules specific to NFC and then applies behavioural features with an Isolation Forest model to further detect anomalies. The last one doesn't just state a card is cloned or malicious, it also gives you a risk score and explanation. The uploaded sample individual report has an abstract, introduction, literature survey, research gap, research problem, objectives, methodology, requirements, budget, work plan and references. This report is presented in a similar format and modified to fit the context of NFC behavioural anomaly detection component.

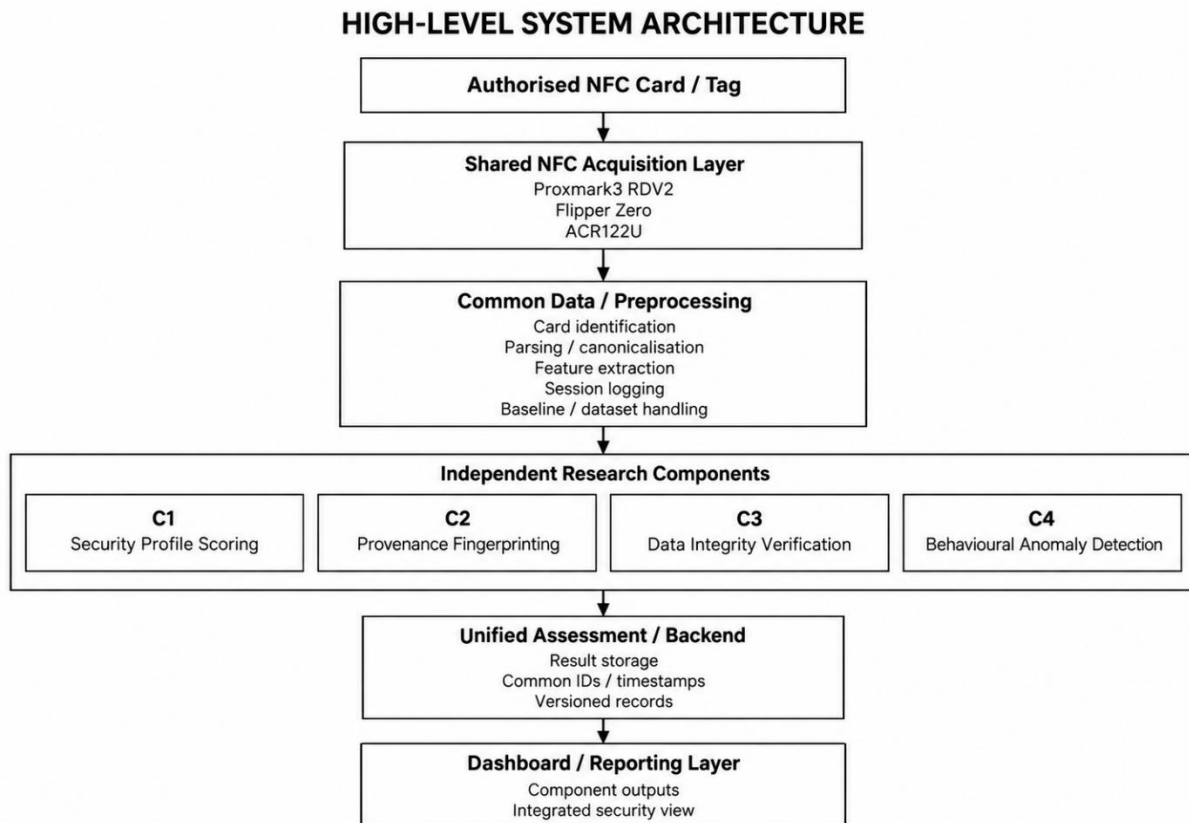
HIGH-LEVEL SYSTEM ARCHITECTURE

Overview

The proposed research introduces an NFC Card and Tag Security Enhancement Framework Using Hybrid Detection that combines four independent but complementary components: C1 - NFC Security Profile Scoring, C2 - Multi-Session Protocol-Response Provenance Fingerprinting, C3 - NFC Tag Data Integrity Verification, and C4 - Hybrid Behavioral Anomaly Detection. Each component addresses a different NFC security problem while contributing to a unified assessment.

The framework begins with authorized NFC cards and tags, with data collected through a shared acquisition layer using devices such as Proxmark3 RDV2, Flipper Zero, and ACR122U. After preprocessing, the collected evidence is provided to the four components in parallel, where each component performs its analysis independently and produces a component-specific result.

Finally, the system stores the outputs in a shared backend and presents them through a unified dashboard and reporting layer. This architecture maintains component independence while providing an integrated view of NFC card and tag security.



6. Background and Literature Survey

6.1 NFC-Based Access and Authentication Systems

NFC systems use short-range wireless communication to identify cards, tags, or compatible devices. In an access control environment, a user presents an NFC credential to a reader. The system reads the credential information, checks the authentication result, and decides whether access should be allowed or denied.

❖ Although this process is useful for authentication, it mainly answers the question:

“Is this credential accepted by the system?”

It does not fully answer the question:

“Is this credential being used in an expected and normal way?”

❖ Behavioural information can include:

- How frequently the credential is scanned
- Where the credential is used
- The time between two scans
- Whether authentication repeatedly fails
- Whether the credential is used during unusual hours
- Whether the same credential appears in different reader locations
- Whether the observed tag family is consistent with previous observations

These behavioural characteristics can be used to identify abnormal credential usage.

6.2 Rule-Based Anomaly Detection

❖ Rule-based detection uses predefined conditions to identify suspicious events. For example:

- More than five scans within ten seconds
- Three or more consecutive authentication failures
- Access outside the permitted time period
- A credential appearing at two distant locations within an unrealistic time gap

❖ Rule-based detection has several advantages:

- It is easy to understand.
- It is easy to implement.
- It provides clear explanations.
- It is suitable for known and predefined suspicious patterns.
- It does not require a large training dataset.

However, rule-based detection also has limitations. It can only detect behaviours that have been explicitly defined. If an abnormal pattern does not match any rule, it may remain undetected. Rules may also generate false positives when legitimate users behave differently from the expected pattern.

6.3 Machine Learning-Based Anomaly Detection

Machine learning-based anomaly detection identifies data points that differ significantly from normal behaviour. Instead of depending only on manually written rules, the model analyses patterns in the data.

For this component, Isolation Forest is proposed. Isolation Forest is an unsupervised anomaly detection algorithm. It isolates observations by repeatedly dividing the feature space. Unusual observations are generally easier to isolate than normal observations.

❖ Isolation Forest is suitable for this component because:

- It can work without a large set of labelled attack examples.
- It can identify unusual combinations of behavioural features.
- It is relatively efficient.
- It can be used with numerical behavioural features.
- It can detect patterns that are not explicitly defined in the rule engine.

For example, a credential may not violate any single rule, but the combination of slightly higher scan frequency, unusual access time, changing reader locations, and increased authentication failures may make the overall behaviour unusual. Isolation Forest can analyse this combination.

6.4 Hybrid Rule–ML Detection

A hybrid detection system combines deterministic rules with machine learning. The rule engine identifies known suspicious patterns, while the machine learning model identifies statistical anomalies and unusual combinations of features.

The main advantages of a hybrid approach are:

- Better coverage of known and unknown anomalies

- More explainable alerts
- Use of domain knowledge through rules
- Ability to identify complex behavioural patterns
- Reduced dependence on only one detection method

In the proposed N4 architecture, the rule engine does not operate separately from the ML model. The results of the rule engine are converted into features, especially the rule flag count, and are supplied to the machine learning stage. This creates a connection between human-defined security knowledge and data-driven anomaly detection.

6.5 Isolation Forest

Isolation Forest was introduced as an anomaly detection method based on the idea that anomalies are easier to isolate than normal observations. The algorithm creates random partitions of the data. Anomalous observations generally require fewer partitions to become isolated.

❖ In this project, Isolation Forest will receive behavioural features such as:

- Scan velocity
- Authentication failure rate
- Location entropy
- Tag-family consistency
- Time-of-day score
- Inter-scan gap standard deviation
- Rule flag count

The model will produce an anomaly score. This score will then be combined with rule evidence to generate a final risk score.

7. Research Gap

Existing NFC security systems mainly focus on credential verification, physical card characteristics, protocol-level behaviour, tag-data integrity, or clone-like detection. These methods are important, but they do not always focus on the behavioural pattern of a credential over time.

❖ The following gaps can be identified:

1. Limited focus on behavioural misuse

Many NFC security solutions focus on whether the credential is genuine or whether the tag data is correct. Less attention is given to how the credential is used over time.

2. Dependence on predefined rules

Rule-based systems can detect known suspicious patterns, but they may fail to detect new or subtle behavioural anomalies.

3. Limited use of NFC-specific behavioural features

General anomaly detection methods may not use NFC-specific information such as scan intervals, reader locations, authentication sequences, and tag-family consistency.

4. Lack of a rule-to-ML connection

In some systems, rules and machine learning operate as separate methods. The rule results are not always used as input features for the ML model.

5. Limited systematic comparison

There is a need to compare Rule-only, ML-only, and Hybrid Rule–ML approaches using the same dataset and evaluation metrics.

6. Limited explainability

Machine learning models may provide an anomaly score without clearly explaining why the behaviour was considered unusual. Security users need understandable explanations.

7. Limited hardware-in-the-loop validation

Purely simulated data may not fully represent real NFC interaction events. Therefore, real authorized NFC interactions should be used to validate the event format and basic behaviour.

Based on these gaps, N4 proposes an NFC-specific hybrid behavioural anomaly detection pipeline using real hardware-based event grounding, controlled simulation, predefined rules, behavioural feature engineering, and Isolation Forest.

8. Research Problem

A valid NFC credential may still be used in an unusual or suspicious manner. Traditional authentication systems generally verify whether the credential is accepted, but they may not identify abnormal usage patterns such as repeated rapid scanning, unrealistic movement between readers, repeated authentication failures, unusual access times, or inconsistent tag-family observations.

Rule-based systems can detect some predefined behaviours, but they may miss unknown or complex anomalies. Machine learning models can identify unusual data patterns, but they may lack explainability and may produce false positives when used without NFC-specific domain knowledge.

Therefore, the research problem of this component is:

How can an NFC system detect unusual credential usage by combining NFC-specific rules and machine learning-based behavioural anomaly detection in an explainable and effective hybrid pipeline?

❖ The proposed solution is a Hybrid Rule–ML Pipeline that:

1. Collects NFC interaction events.
2. Applies NFC-specific behavioural rules.
3. Extracts behavioural features.
4. Uses Isolation Forest to detect statistical anomalies.
5. Combines rule evidence and ML results.
6. Produces a risk score, alert level, and explanation.

9. Objectives

9.1 Main Objective

To design and evaluate an NFC-specific hybrid behavioural anomaly detection system that combines predefined rules and Isolation Forest machine learning to identify unusual NFC credential usage.

9.2 Sub Objectives

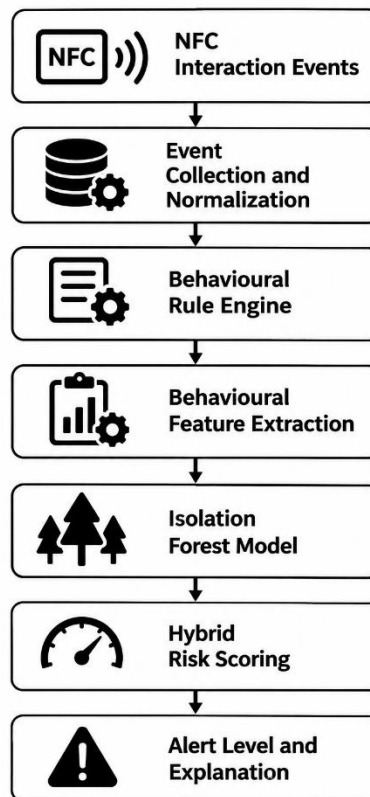
The sub-objectives of this component are:

1. To collect NFC interaction events from authorized hardware experiments and controlled simulation.
2. To design a normalized event format for NFC behavioural analysis.
3. To develop NFC-specific rules for detecting known suspicious behaviours.
4. To extract meaningful behavioural features from NFC event sequences.
5. To implement an Isolation Forest model for unsupervised anomaly detection.
6. To use rule results as input features for the machine learning stage.
7. To generate a combined risk score and alert level.
8. To provide human-readable explanations for detected anomalies.
9. To compare Rule-only, ML-only, and Hybrid Rule–ML approaches.
10. To evaluate the system using precision, recall, F1-score, false-positive rate, PR-AUC, detection latency, and risk calibration.
11. To validate the system using real authorized NFC interactions and controlled simulated scenarios.
12. To identify the limitations and possible improvements of the proposed approach.

10. Proposed Methodology

10.1 Overview of the Proposed Architecture

❖ The proposed N4 system follows the architecture below:



The system uses two main detection stages:

- Stage 1 – Rule-Based Detection
- Stage 2 – Machine Learning-Based Detection

The rule engine identifies known suspicious patterns. The machine learning model identifies unusual behavioural combinations that may not be covered by the rules.

10.2 NFC Event Collection

The first stage is collecting NFC interaction events.

The event data may contain the following fields:

Field	Description
event_id	Unique identifier for the event
credential_alias	Anonymous credential name such as TAG001
timestamp	Date and time of the event
reader_id	Identifier of the NFC reader
location	Logical location of the reader
authentication_result	Success or failure
card_family	Observed tag family
event_type	Type of NFC interaction

For privacy and security reasons, real personal information should not be stored. Anonymous aliases such as TAG001 and TAG002 can be used instead of real card identifiers.

Real authorized NFC interactions can be collected using available NFC hardware. These interactions help validate the structure of the event data. Controlled simulation can then be used to create larger and more diverse behavioural scenarios.

Simulation is necessary because two physical NTAG215 cards alone cannot produce enough different behavioural patterns for meaningful machine learning evaluation. Some scenarios, such as tag-family mismatch, may also require controlled simulated data or an authorized tag from another family.

10.3 Data Preprocessing and Normalization

The collected events will be cleaned and prepared before applying the rules and machine learning model.

The preprocessing activities include:

❖ Data Cleaning

- Remove duplicate events.
- Identify missing values.
- Correct inconsistent timestamps.
- Check invalid reader identifiers.
- Check invalid authentication results.
- Remove unnecessary personal information.

❖ Timestamp Processing

Timestamps will be used to calculate:

- Time between consecutive scans
- Number of scans within a time window
- Access hour
- Day or night status
- Time gap between different reader locations

❖ Event Ordering

Events belonging to the same credential will be arranged chronologically. This is important for identifying sequences such as:

- Repeated authentication failures
- Rapid scanning
- Movement between readers
- Changes in access behaviour

❖ Data Encoding

Categorical values such as reader IDs, locations, authentication results, and card families will be converted into a format suitable for analysis.

❖ **Data Partitioning**

The dataset will be divided into:

- Development or training data
- Validation data
- Test data

The test data should contain both normal and controlled anomalous scenarios.

10.4 Behavioural Rule Engine

The rule engine will contain five NFC-specific rules.

Rule 1 – Scan Burst

This rule detects when the same credential is scanned more than five times within ten seconds.

IF same credential scans > 5 times within 10 seconds

THEN trigger R1

This may indicate repeated scanning, accidental repeated presentation, or suspicious automated activity.

Rule 2 – Impossible Travel

This rule detects when the same credential appears at different reader locations within an unrealistic time gap.

IF the same credential appears at two different locations

within an unrealistic time interval

THEN trigger R2

For example, a credential appearing at the Main Gate and then at a distant Laboratory reader within a few seconds may be considered anomalous.

Rule 3 – Authentication Failure Sequence

This rule detects three or more consecutive authentication failures.

IF authentication failures ≥ 3 consecutively

THEN trigger R3

Repeated failures may indicate incorrect use, attempted unauthorized access, or abnormal interaction behaviour.

Rule 4 – Tag-Family Mismatch

This rule checks whether the observed tag family is inconsistent with the expected tag family of the credential.

IF observed tag family differs from expected tag family

THEN trigger R4

This rule should be interpreted carefully. A tag-family mismatch does not automatically prove cloning or malicious activity. It only indicates that the observed tag information is inconsistent with the expected profile.

Rule 5 – Unusual Access Time

This rule detects access during time periods that are outside the expected access schedule.

IF credential is used outside its normal access time

THEN trigger R5

For example, a credential normally used during working hours may be flagged if it is used late at night, depending on the defined access policy.

10.5 Behavioural Feature Engineering

After the rule engine processes the events, the system will extract behavioural features.

The proposed seven features are shown below.

Feature	Description
Scan velocity	Number of scans within a specific time period
Authentication failure rate	Failed authentication attempts divided by total attempts

Location entropy	Diversity or unpredictability of locations used by a credential
Tag-family consistency	Level of consistency between observed and expected tag families
Time-of-day score	Degree to which access time differs from the normal schedule
Inter-scan gap standard deviation	Variation in the time intervals between consecutive scans
Rule flag count	Number of rules triggered for the credential or event window

❖ Scan Velocity

Scan velocity measures how frequently a credential is scanned.

A high scan velocity may indicate:

- Repeated scanning
- Automated scanning
- Accidental repeated presentation
- Suspicious rapid access attempts

❖ Authentication Failure Rate

This feature is calculated as:

$$\text{Authentication Failure Rate} = \frac{\text{Number of Failed Attempts}}{\text{Total Authentication Attempts}}$$

A high failure rate may indicate abnormal access behavior.

❖ **Location Entropy**

Location entropy measures how diverse the credential's reader locations are.

A credential that normally appears at one or two locations may become unusual if it suddenly appears at many different locations within a short period.

❖ **Tag-Family Consistency**

This feature measures whether the observed tag family matches the expected tag family.

A consistent tag family produces a normal value. An inconsistent observation may increase the anomaly level.

❖ **Time-of-Day Score**

This feature measures how far the access time is from the expected access schedule.

For example, access during an approved working period may receive a normal score, while access during an unusual period may receive a higher anomaly score.

❖ **Inter-Scan Gap Standard Deviation**

This feature measures how much the time gap between consecutive scans changes.

Unusual variation in scan intervals may indicate abnormal or automated behaviour.

❖ **Rule Flag Count**

Rule flag count represents the number of predefined rules triggered by the event or behavioural window.

This is an important feature because it acts as a bridge between the rule engine and the machine learning model.

For example,

R1 triggered = 1

R2 triggered = 0

R3 triggered = 1

R4 triggered = 0

R5 triggered = 1

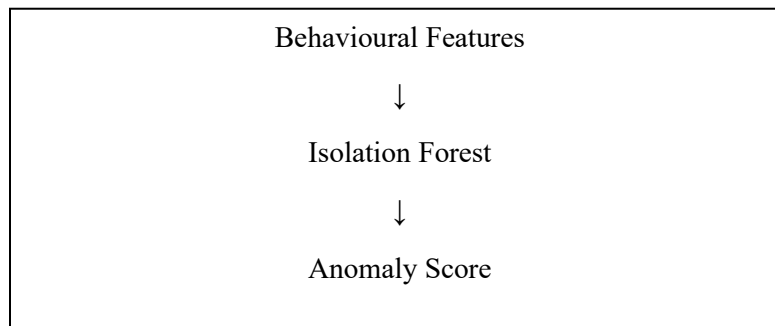
Rule flag count = 3

10.6 Isolation Forest Model

The machine learning stage will use Isolation Forest.

Isolation Forest is an unsupervised anomaly detection algorithm. It identifies unusual observations by isolating them from the rest of the data.

The model will receive the behavioural features extracted from the NFC events.



The model can identify:

- Unusual combinations of normal-looking features
- New behavioural patterns
- Subtle anomalies
- Patterns not directly covered by the rules

For example, an event may not violate the scan burst rule or authentication failure rule individually. However, the combination of unusual access time, high location diversity, and irregular scan intervals may still be identified as anomalous by Isolation Forest.

The model will be trained mainly using normal or mostly normal behavioural data. Controlled anomalous scenarios will be used during testing.

10.7 Hybrid Rule–ML Pipeline

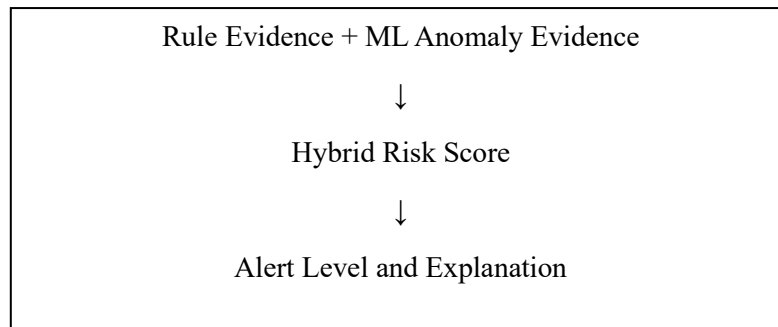
The hybrid pipeline combines the outputs of the rule engine and the Isolation Forest model.

The process is:

1. Collect NFC events.
2. Normalize and order the events.
3. Apply the five predefined rules.

4. Extract behavioural features.
5. Include rule flag count as a bridge feature.
6. Apply Isolation Forest.
7. Combine rule evidence and ML anomaly score.
8. Generate the final risk score.
9. Produce an alert level and explanation.

❖ The proposed logic can be represented as:



The hybrid approach is expected to provide better detection coverage than using only rules or only machine learning.

10.8 Risk Scoring and Alert Generation

The final system will generate a risk score between 0 and 100.

An example risk classification is:

Risk Score	Alert Level	Meaning
0–29	Low	Behaviour is close to the expected pattern
30–59	Medium	Some unusual behaviour is detected

60–79	High	Multiple suspicious indicators are present
80–100	Critical	Strong behavioural anomaly evidence is present

The risk score will be based on:

- Number of triggered rules
- Severity of the triggered rules
- Isolation Forest anomaly score
- Behavioural feature values
- Historical behaviour of the credential

The output should include a human-readable explanation.

Example,

Credential: TAG001

Risk Score: 78

Alert Level: High

❖ Reasons:

- More than five scans within ten seconds
- Three consecutive authentication failures
- Access occurred outside the normal time period

The system should not state that the card is definitely cloned, stolen, or malicious. A high risk score only means that the observed usage is unusual according to the defined rules and model.

10.9 Evaluation and Validation

The proposed system will be evaluated using three detection approaches.

❖ Rule-Only Approach

Only the predefined rules will be used.

This approach will show how well known suspicious behaviours can be detected without machine learning.

❖ ML-Only Approach

Only the Isolation Forest model will be used.

This approach will show whether the machine learning model can identify unusual behavioural patterns without explicit rule evidence.

❖ Hybrid Rule–ML Approach

Both the rule engine and Isolation Forest will be used.

This is the proposed N4 approach.

❖ Evaluation Metrics

The following metrics will be used,

Precision

Precision measures how many detected anomalies are actually anomalous.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

Recall measures how many actual anomalies are correctly detected.

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score

F1-score combines precision and recall.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

False Positive Rate

False positive rate measures how many normal events are incorrectly flagged.

$$\text{FPR} = \frac{FP}{FP + TN}$$

PR-AUC

PR-AUC measures the relationship between precision and recall across different decision thresholds. It is useful when anomalous events are less common than normal events.

Detection Latency

Detection latency measures the time taken from event collection to alert generation.

Risk Calibration

Risk calibration checks whether the generated risk scores properly represent the level of anomaly evidence.

10.10 Additional Validation Tests

The following tests will also be conducted.

❖ Bridge Feature Ablation

The rule `flag_count` feature will be removed and the results will be compared with the complete hybrid model. This will show whether the rule-to-ML bridge improves performance.

❖ Unseen Anomaly Test

One type of anomaly will be excluded during development and introduced only during testing. This will show whether the model can generalize to previously unseen patterns.

❖ **Cross-Reader Validation**

The model will be developed using events from some readers and tested using events from another reader. This will show whether the system is robust across different reader locations.

❖ **False-Positive Analysis**

Normal events that are incorrectly flagged will be examined. This will help identify whether the rules or model are too sensitive.

11. Final Output

The final output of Novelty 4 will be a prototype NFC behavioural anomaly detection component.

The expected output includes:

1. NFC event collection module
2. Normalized NFC event dataset
3. Behavioural rule engine
4. Behavioural feature extraction module
5. Isolation Forest anomaly detection model
6. Hybrid risk-scoring mechanism
7. Alert level classification
8. Human-readable explanation
9. Evaluation results comparing three approaches
10. Documentation of limitations and future improvements

result,

Credential Alias: TAG002

Risk Score: 84

Alert Level: Critical

Triggered Rules,

- R1: Scan burst
- R2: Impossible travel
- R5: Unusual access time

ML Result,

Behavioural pattern identified as highly unusual

Final Explanation:

The credential shows rapid scanning, unrealistic movement between reader locations, and access outside the expected time.

The component will support the overall NFC security project by focusing on behavioural misuse and anomalous usage patterns.

12. Project Requirements

12.1 The system shall,

1. Collect NFC interaction events from authorized hardware or controlled simulation.
2. Store event information using an anonymous credential alias.
3. Record timestamps, reader IDs, locations, authentication results, and tag-family information.
4. Normalize and clean the collected data.
5. Order events chronologically for each credential.
6. Implement the five predefined behavioural rules.
7. Calculate the seven behavioural features.
8. Calculate the rule flag count.

9. Train and execute the Isolation Forest model.
10. Combine rule evidence and ML anomaly evidence.
11. Generate a risk score between 0 and 100.
12. Classify the risk score into alert levels.
13. Display the triggered rules.
14. Generate a human-readable explanation.
15. Store detection results for later evaluation.
16. Compare Rule-only, ML-only, and Hybrid approaches.
17. Support controlled simulation of behavioural scenarios.
18. Provide evaluation metrics such as precision, recall, F1-score, and false-positive rate.

12.2 Non-Functional Requirements

❖ Accuracy

The system should detect known and unknown behavioural anomalies with acceptable precision and recall.

❖ Explainability

Every alert should provide understandable reasons, including triggered rules and important behavioural indicators.

❖ Performance

The system should process NFC events with low detection latency.

❖ Privacy

Real personal information and unnecessary card identifiers should not be stored. Anonymous aliases should be used.

❖ Reliability

The system should continue processing events even when individual events contain missing or inconsistent values.

❖ Scalability

The design should support multiple credentials and reader locations.

❖ Maintainability

Rules, thresholds, features, and model parameters should be easy to update.

❖ Usability

The final output should be understandable to a security administrator or project evaluator.

❖ Security

The event logs and model outputs should be protected from unauthorized modification.

13. System Requirements

13.1 Hardware Requirements

Hardware	Purpose
NFC-compatible cards/tags	Generate authorized NFC interactions
NTAG215 cards	Physical event generation and validation
Flipper Zero	Secondary NFC interaction, validation, and demonstration
Proxmark3, if available	Primary NFC research and measurement device
NFC reader	Read NFC credentials and generate events
Computer or laptop	Data processing, model development, and testing
USB cable	Connect hardware devices

Storage device	Store datasets and experiment results
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The Flipper Zero will be used only for authorized interaction, validation, and demonstration. It is not itself the anomaly detection model. The anomaly detection is performed by the software pipeline.

13.2 Software Requirements

Software	Purpose
Python 3.x	Main development language
Pandas	Data processing
NumPy	Numerical calculations
Scikit-learn	Isolation Forest implementation
Matplotlib	Data visualization
Jupyter Notebook or Google Colab	Experimentation and model development
SQLite or CSV	Local event storage

Visual Studio Code or PyCharm	Code development
Git	Version control
Operating System	Windows or Linux

14. Component Budget

The budget depends on the hardware already available to the researcher. Since the user already has two NTAG215 cards and a Flipper Zero, the additional cost may be limited.

Item	Category	Estimated Cost (LKR)	C2-Specific Justification
Proxmark3	Research Hardware	25,000	Primary reader for repeated protocol-response and timing acquisition.
Flipper Zero	Research Hardware	80,000	Authorized emulation source for genuine-vsimulated comparison.
NFC Card Set	Test Media / Consumables	10,000	Multiple authorized card families for repeatable C2 experiments.
Python 3.x	Software	Free	Data processing, feature extraction, and model development.
Pandas / NumPy	Software	Free	Cleans and summarizes repeated session measurements.
scikit-learn	Software	Free	Implements and evaluates RF, SVM, and KNN models.

Jupyter Notebook	Software	Free	Exploratory analysis and model evaluation.
Proxmark3 Client / Iceman Firmware	Software	Free	Controls Proxmark3 and supports NFC data acquisition.
Git / GitHub	Development Platform	Free	Version control and experiment tracking.
Human Effort	Research Effort	Not monetized	Student-led data collection, analysis, and documentation.

The component mainly uses open-source software. Therefore, the main cost is related to NFC hardware and computing equipment.

15. Work Plan

Activity	Description	Expected Output
Week 1	Study NFC behavioural anomaly detection	Literature notes
Week 2	Design event format	Event schema
Week 3	Collect authorized NFC events	Initial dataset
Week 4	Create controlled simulation scenarios	Extended dataset
Week 5	Implement preprocessing	Clean dataset
Week 6	Implement five behavioural rules	Rule engine
Week 7	Develop behavioural features	Feature dataset
Week 8	Implement Isolation Forest	ML model
Week 9	Develop hybrid risk scoring	Hybrid pipeline
Week 10	Generate alerts and explanations	Output module
Week 11	Evaluate Rule-only, ML-only, and Hybrid	Evaluation results
Week 12	Conduct ablation and validation tests	Validation report
Week 13	Preparing documentation and presentation	Final report and slides

16. Limitations and Ethical Considerations

❖ Limited Physical Dataset

Only a small number of physical NFC cards may be available. Therefore, real hardware data alone may not provide enough diversity for machine learning. Controlled simulation will be used to supplement physical events.

❖ Limited Tag-Family Diversity

If all available cards belong to the NTAG215 family, a genuine physical tag-family mismatch cannot be fully demonstrated. Therefore, the tag-family mismatch scenario may need to be simulated or tested using an authorized tag from another family.

❖ Behavioral Anomaly Is Not Proof of Malicious Intent

An unusual event does not always mean that the user is attacking the system. Legitimate situations, such as emergency access, system testing, or a change in work schedule, may also create unusual patterns.

Therefore, the system should produce a risk indication rather than a final accusation.

❖ Privacy

Real card UIDs, personal names, student numbers, and other identifying information should not be included in the dataset. Anonymous aliases should be used.

❖ Controlled Testing

All NFC experiments must be conducted only on authorized cards, readers, and systems. The project should not involve unauthorized access, credential misuse, or testing against third-party systems.

❖ Model Limitations

Isolation Forest may produce false positives when normal behaviour changes. The model should therefore be evaluated regularly and updated when the system's normal usage patterns change.

17. Conclusion

Novelty 4 proposes an NFC Behavioural Anomaly Detection system using a Hybrid Rule–ML Pipeline. The component focuses on identifying unusual credential usage rather than directly proving that a card is cloned, stolen, or malicious.

The proposed system combines five NFC-specific behavioural rules with seven behavioural features and an Isolation Forest model. The rule engine detects known suspicious patterns such as scan bursts, impossible travel, authentication failure sequences, tag-family mismatch, and unusual access times. The machine learning model identifies unusual combinations of behavioural features that may not be detected by predefined rules. The rule_flag_count feature connects the rule engine with the machine learning stage and forms an important part of the proposed architecture. The final system produces a risk score, alert level, triggered rules, and a human-readable explanation.

The component will be evaluated using Rule-only, ML-only, and Hybrid approaches. Additional tests such as bridge-feature ablation, unseen anomaly testing, cross-reader validation, and false-positive analysis will be used to examine the effectiveness and reliability of the system.

18. References

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Thank you